Extension: Geometric mean

1 Find the geometric mean of the following set of numbers, in exact form:

a 45 and 5

b $6, 10, 60, 75,$ and 90

2 Find the geometric mean of the following set of numbers, rounded to two decimal places:

a 19 and 13

b $3, 4, 12, 13,$ and 15

3 Find the side length of the following squares, rounding to three decimal places, if necessary:

a A square with the same area as a rectangle with length 100 m and width 4 m.

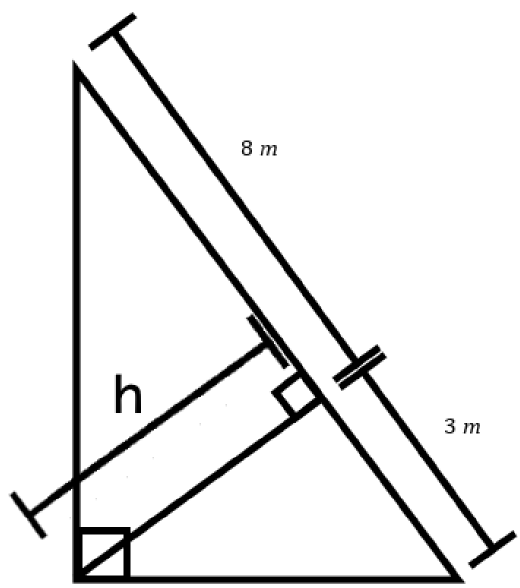
b A square with the same area as a rectangle with length 29 mm and width 11 mm.

4 Find the side length of the following cubes, rounding to four decimal places, if necessary:

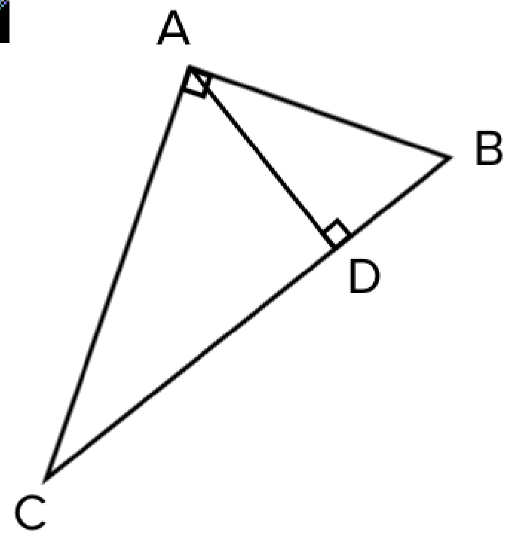
a A cube with the same volume of a rectangular prism with length 21 mm , width 9 mm, and height 49 mm.

b A cube with the same volume of a rectangular prism with length 19 m, width 11 m, and height 29 m.

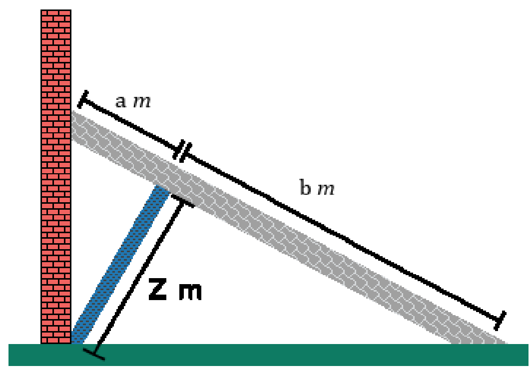
5 Find the exact value of h in the following triangle:



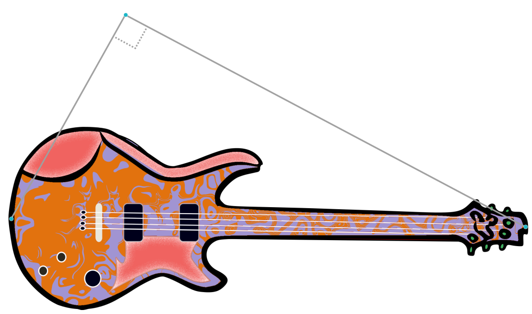
6 In the following diagram, the length of $\overline{AB}$ is 8 units and the length of $\overline{BD}$ is 3 units. Find the length of $\overline{CB}$, rounded to one decimal place.



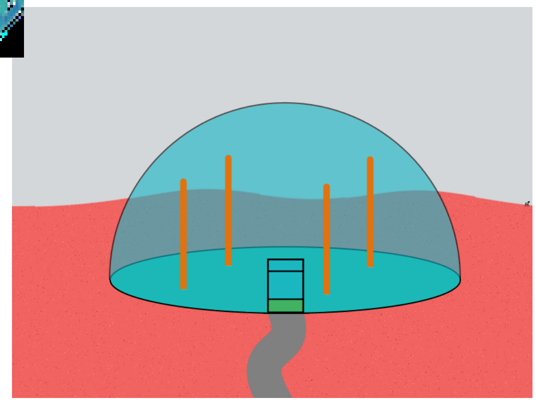
7 An engineer needs to find the length of a beam to support the beam holding the wall. Assume that the two beams are perpendicular to one another, and $a = 4$ and $b = 1$. Find z, rounded to one decimal place.



8 A famous guitarist hangs a guitar exactly 40 cm from the roof of her studio. The guitar hangs horizontally, with ropes securing it to the ceiling attached at either end of the guitar.
The center of mass of the guitar hangs directly below the point where the two ropes meet. If the distance from the center of mass to one end of the guitar is 80 cm, how long is the guitar? Round your answer to the nearest whole number.



9 A proposal for a Martian colony includes a design for hemispherical domes. Each dome has a radius of $54 \mathrm{~m}$, and is supported by four vertical pillars placed $27 \mathrm{~m}$ from the edge of the dome.
How tall must the pillars be to reach the dome? Give your answer as a simplified radical.



10 A $\$100$ investment in the company RocketZ grows by 40% in the first year, 10% in the second year, and 30% in the third year. The same investment in the company Lesta grows by $p\%$ every year. At the end of 3 years the value of investments in RocketZ and Lesta are exactly the same.

a Find the value of p to one decimal place.

b Let $a\%$, $b\%$ and $c\%$ be the interest rates for three consecutive years on an initial investment of $\$I$.
What information does the geometric mean $\sqrt[3]{(100+a)(100+b)(100+c)}$ provide?

11 Applicants for a position at a retail company are given three scores. Their Academic score is out of 100, their Impression score is out of 30 and their Reference score is out of 5.
The latest three applicants have their scores given in the table.

Name	Academic	Impression	Reference
Beth	94	14	1
Isabelle	70	27	4
Ursula	73	29	3

a What is the arithmetic mean of Beth's scores? Give your answer to two decimal places.

b What is the geometric mean of Beth's scores? Give your answer to two decimal places.

c Complete the table to two decimal places.

Applicant	Arithmetic mean	Geometric mean
Beth	36.33	10.96
Isabelle		
Ursula		

d The retail company wants to hire the candidate that did the best overall, giving equal weight to the proportion of each score. Which applicant should the company hire?

12 Two town councils measure the amount of pollutants (in ppm) in their water supply every week. The results from the last six weeks are shown in the following table:

	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6
Town 1	1.5	2.2	2	2.8	54	2
Town 2	9	9	9	9	9	9

a Complete the table to the nearest whole number.

	Arithmetic Mean	Geometric Mean
Town 1		
Town 2		

b Let x be the next reading of Town 2's water supply. Find the value of x for Town 2's arithmetic mean to reach the same level as Town 1's arithmetic mean for the last six weeks.

c Let y be the next reading of Town 1's water supply. Find the value of y for Town 1's geometric mean to reach the same level as Town 2's geometric mean for the last six weeks. Round your answer to the nearest whole number.

13 Lisa begins with a rectangle that has length 5 cm and width 30 cm . Using this information, she makes two squares: One from cardboard and one from metal.
The cardboard square has the same perimeter as the rectangle, while the metal square has the same area.

a Determine the side length (in simplest form) of the following:

i Cardboard square

ii Metal square

b Lisa looks at many more rectangles of varying dimensions, creating a cardboard and a metal square for each one in the same way as above. State whether each of the following is true.

i The cardboard square is always larger or equal to the metal square.

ii Sometimes the metal square is larger, and sometimes the cardboard square is larger.

iii If the original rectangle is not a square, then the metal square and the cardboard square will be different sizes.

iv The metal square is always larger or equal to the cardboard square.

c If Lisa began with a rectangle that has le[redacted] cm and width $w\ \text{cm}$, find the side length of the following:

i Cardboard square

ii Metal square

d Determine whether each of the following is always true for a rectangle with length $l\ \text{cm}$ and width $w\ \text{cm}$.

i $\sqrt{lw} \leq \dfrac{l+w}{2}$

ii $\sqrt{lw} = \dfrac{l+w}{2}$

iii $\sqrt{lw} > \dfrac{l+w}{2}$

iv $\sqrt{lw} \geq \dfrac{l+w}{2}$